

VIKAS PAL

Software Engineer | AI/ML Enthusiast | Open Source Contributor
vikas83pal@gmail.com | +91 6302956492 | Hyderabad, Telangana, India
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

INTRODUCTION

A passionate Software Engineer and AI/ML enthusiast with hands-on experience in building scalable applications, optimizing machine learning models, and contributing to open-source projects. Skilled in Python, Java, React, Spring Boot, and Docker, with a strong foundation in problem-solving and software development. Actively seeking opportunities to leverage my skills in a challenging environment.

EDUCATION

J.N.T.U.H University College of Engineering Sultanpur <i>B.Tech in Computer Science and Engineering CGPA: 7.90</i>	Sultanpur, Telangana 2023 - Present
Quli Qutub Shah Govt Polytechnic College <i>Diploma in Computer Science and Engineering CGPA: 9.37</i>	Hyderabad, Telangana 2020 - 2023

EXPERIENCE

Intern IIT Hyderabad	Hyderabad Feb 2025
- Built ML models for CodeKernel using graph kernels to optimize API usage example selection and automated code suggestions. - Improved model accuracy by 15% through hyperparameter tuning and feature engineering.	
Research Intern IIT Hyderabad	Hyderabad Sep 2024
- Developed autonomous navigation for UGVs using A*, SLAM, and real-time obstacle detection. - Implemented path-planning algorithms to reduce navigation time by 20%.	
Contributor Google Summer of Code (GSoC)	Hyderabad Jan 2023 - Nov 2023
- Improved UI performance in an open-source project using React and optimized API calls. - Reduced page load time by 30% through code optimization and lazy loading.	
Industrial Training Java & Spring Boot Developer	Hyderabad June 2022 - Nov 2022
- Built RESTful APIs, integrated JWT authentication, and implemented microservices using Spring Boot, Hibernate, and Docker. - Reduced API response time by 25% through caching and database optimization.	

SKILLS

Programming Languages:	C, C++, Java, Python, JavaScript
Libraries/Frameworks:	React, Spring Boot, WebSocket, NumPy, Pandas
Tools/Platforms:	Git, GitHub, MERN, VS Code, Linux, Postman, Figma, Docker, Kubernetes, Jenkins, Apache Kafka
Databases:	SQL, MongoDB

PROJECTS

JNTU Result Analyzer Link	<i>React, Redux, REST API</i>
- Developed a web application to analyze and visualize student academic performance using React and Redux for state management. - Integrated RESTful APIs to fetch and process data from the JNTU database, enabling real-time updates and detailed performance reports. - Reduced manual effort by 80% by automating the generation of performance summaries and trend analysis.	
Women Safety Application Link	<i>Java, WebSocket</i>
- Designed and implemented a real-time emergency alert system using WebSocket for instant communication between users and emergency contacts. - Integrated geolocation tracking and Firebase for secure authentication and real-time data storage. - Enhanced user safety by providing real-time location updates and SOS alerts with a response time of less than 2 seconds.	
Brain Tumor Detection Link	<i>Python, YOLO</i>
- Trained a YOLO-based deep learning model to detect brain tumors in MRI scans with 90% accuracy. - Implemented data preprocessing techniques such as normalization and augmentation to improve model performance.	

- Deployed the model as a web application using Flask, enabling doctors to upload and analyze MRI scans in real-time.

Real-Time Chat Application | [Link](#)

Spring Boot, React, WebSocket

- Built a real-time chat application supporting multi-room communication, user authentication, and message history.
- Utilized WebSocket for bidirectional communication and React for a responsive and intuitive user interface.
- Achieved a message delivery latency of less than 100ms, ensuring seamless communication.

CERTIFICATIONS

- Java Foundation - **Oracle**
- Problem Solving - **HackerRank**
- Python - **SoloLearn**
- AI/ML for Geodata Analysis - **ISRO**
- Digital Image Processing - **NPTEL**

ACHIEVEMENTS

- First prize in state-level Project Expo at Surjana Tech Fest.
- Google Summer of Code 2023 contributor.
- Contributed to multiple open-source projects.
- Solved 300+ problems on LeetCode, achieving Institute Rank #3 on GeeksforGeeks.
- Published journal article on women's safety solutions.